EE542 Lab 9: A Custom Chatbot with an LLM and Embeddings

Huacong Tang
University of Southern California

Xiao Yang
University of Southern California

Jiachen Lu
University of Southern California

Code Demo Video

Part 1: Group chat with a self-hosted LLM

The template has the web client, SQL schema and Android project but no server, so we wrote one against the API the client calls: FastAPI on MySQL (`users`, `messages`), bcrypt passwords and JSON Web Tokens, and a WebSocket that pushes every message to all members (Figure 1). A message containing "?" is answered in the background by Llama 3.1 8B Instruct (Q4_K_M, 4.9 GB), built with llama.cpp on a MacBook Pro M4 Pro, 24 GB: 41 tokens/s, an answer about 2 s after the question. The suggested 70 B model needs about 40 GB and does not fit.

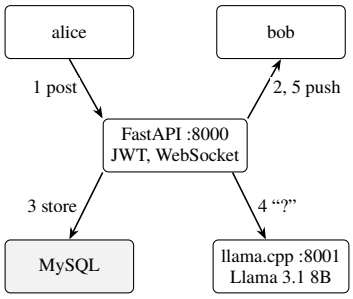


Figure 1: A message is stored and pushed to all members at once; a "?" sends the last 10 messages to the LLM and its answer is pushed as the bot (5).

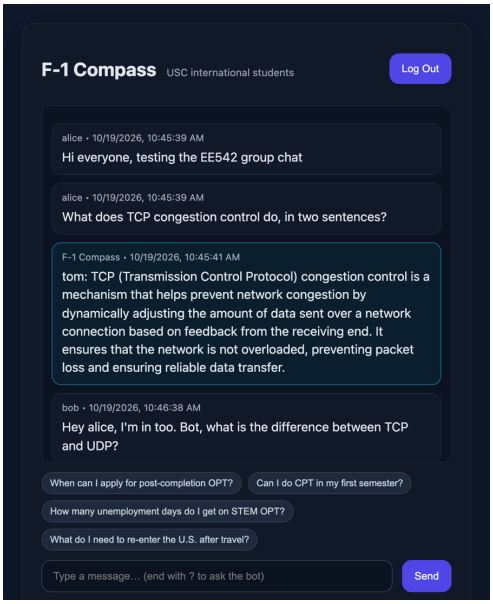


Figure 2: The chat in a browser. The first answer starts with an invented "tom:", copied from the "name: text" history; the prompt now forbids it and the server strips a leading member name.

Tests: two browsers saw each other's messages and the bot's answers live (Figure 2); a wrong password gives 401, a duplicate user name 400, no token 401.

Part 2: The Android app

The app is a Trusted Web Activity: one activity that asks Chrome to show the chat site full screen. Chrome drops the address bar only if the site serves `/.well-known/assetlinks.json` with the app's package name (`edu.usc.ee542.group6.f1compass`) and its signing-key fingerprint, which our server does. We regenerated the template's Windows-only Gradle wrapper; the debug APK is 519 KB. On an Android 14 emulator (Pixel 7, Chrome 113) we reached the laptop with `adb reverse` and, since Chrome checks asset links only over HTTPS, used its developer switch that skips the check for one origin. The app opened full screen (Figure 3). The phone exposed three client bugs, now fixed: side-by-side sign-in fields overflowed, an empty password showed "[object Object]", and the chat opened at the oldest message.

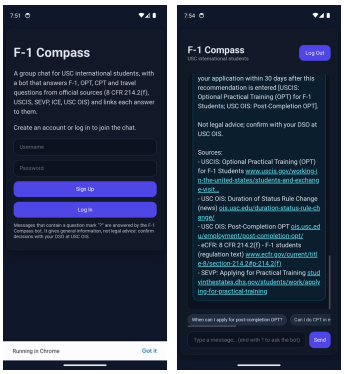


Figure 3: The Android app on the emulator: sign-in (left) and an answer with its sources (right). No address bar.

Part 3: Answering from documents

Figure 4 shows the pipeline. Chunks keep their file and page for citations, and a LangChain `ConversationalRetrievalChain` rewrites a follow-up ("And where do I put the license file?") into a standalone question. It runs as a command-line loop and as a web page that uploads files and shows each retrieved chunk with its similarity and embedding values (Figure 5).

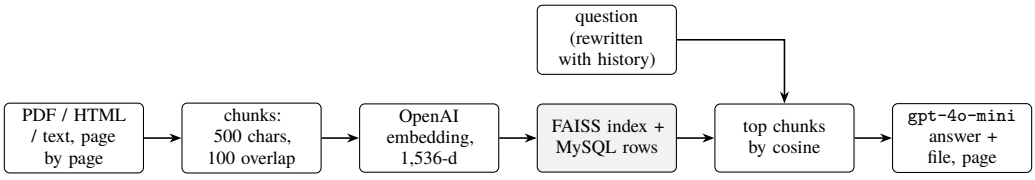


Figure 4: Retrieval-augmented answering.

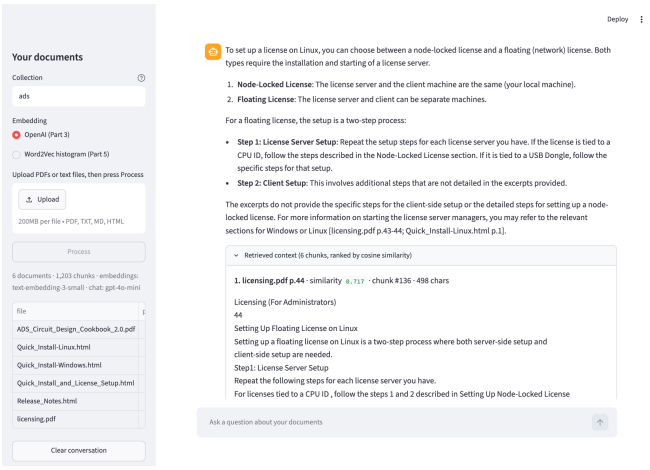


Figure 5: The web page on the six ADS documents (1,203 chunks): the answer cites file and page; below it, the retrieved chunks with similarity.

On the six Keysight ADS documents (two PDFs of 124 and 87 pages, four HTML pages), "How do I install ADS on Linux?" retrieves a chunk at similarity 0.71 and the answer lists the install commands; "What is the capital of France?" finds nothing above 0.16 and is declined. A first prompt that said "answer from the text only" with 4 chunks refused two answerable questions; asking it to answer when anything relevant is retrieved, say what is missing, and refuse only otherwise, with 6 chunks, fixed both.

Part 4: F-1 Compass, a custom chatbot

F-1 students keep asking the same questions (when to file for OPT, whether an internship needs CPT, how many unemployment days are allowed), and the answers are spread over the federal regulation, two agencies and USC's international office. F-1 Compass is the group chat of Part 1, in the browser and the app of Part 2, answering with the pipeline of Part 3 only from those official pages, with links (Figure 6).

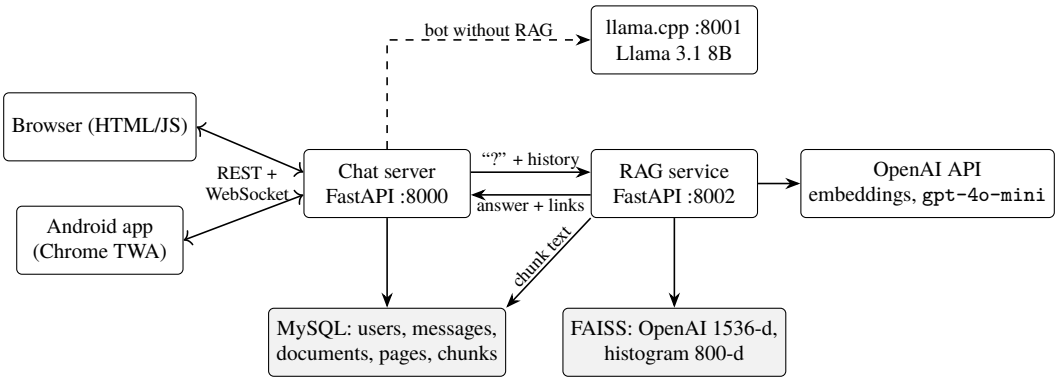


Figure 6: F-1 Compass: Parts 1–3 combined. A "?" message goes to the RAG service, which answers from the F-1 index with source links; without it, the server asks Llama directly (dashed).

Knowledge base. 59 official pages (Figure 7a), main content only, 1,763 chunks; a manifest records each URL and hash so the corpus can be re-downloaded. The regulation stores a paragraph as one line of thousands of characters, so long lines are first broken at sentence ends.

Answering. The prompt asks for the general rule with exact numbers first, federal or USC, each fact cited by agency and page; unrelated questions are declined; every answer ends "Not legal advice;

confirm with your DSO at USC OIS." Retrieval uses maximal marginal relevance (8 of the 30 closest chunks, preferring unlike ones). With plain top-6, a STEM OPT question pulled three chunks of one OIS worked example and the bot gave that student's dates as the rule.

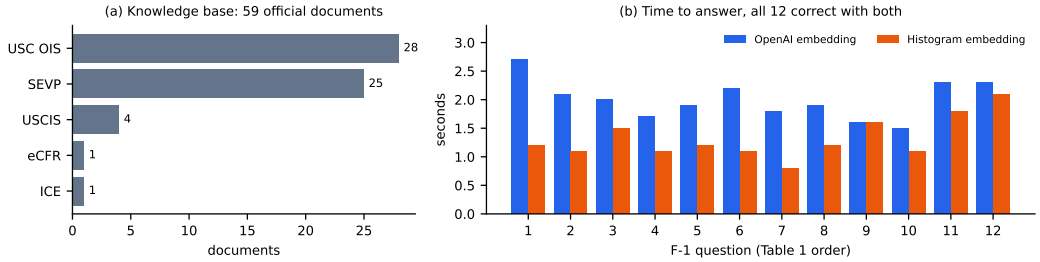


Figure 7: (a) Sources of the F-1 knowledge base. (b) Time from question to answer for the 12 F-1 questions of Table 1, with the two embeddings of Part 5.

Evaluation. Twelve real questions, each with a fact the answer must state, plus one unrelated question (Table 1). A pass needs the fact, a citation of an agency that holds it, and the disclaimer; the unrelated one must be declined. All 13 pass with both embeddings (Figure 7b).

Question	Required fact	OpenAI	Hist.
When can I apply for post-completion OPT?	90 days before / 60 days after	✓	✓
How many hours a week can I work on campus while classes are in session?	20 hours a week	✓	✓
Can I do CPT in my first semester as a graduate student?	one academic year	✓	✓
If I do 12 months of full-time CPT, can I still get OPT?	no OPT after 12 months	✓	✓
How many days of unemployment are allowed during post-completion OPT?	90 days	✓	✓
How many days of unemployment do I get in total with the STEM OPT extension?	150 days	✓	✓
How long is the STEM OPT extension?	24 months	✓	✓
When should I apply for the STEM OPT extension?	90 days before EAD ends	✓	✓
What form does my employer and I complete for a STEM OPT training plan?	Form I-983	✓	✓
How long is the grace period after I complete my program?	60 days	✓	✓
What do I need to re-enter the U.S. after traveling abroad as an F-1 student?	I-20, passport, visa	✓	✓
Can I take a reduced course load because of a medical condition, and for how long?	12 months	✓	✓
What is the best pizza place near the USC campus?	declined	✓	✓

Table 1: Answer test of F-1 Compass. "Required fact" is what the answer must state; a pass also requires a citation of an agency whose page holds the fact and the disclaimer. Last row: an unrelated question, which must be declined.

Part 5: Word-cluster histogram embeddings

Figure 8 shows the embedding: Word2Vec (skip-gram, $N = 100$) trained on the corpus, all word vectors clustered by our own k-means into K clusters, and each chunk described by the share of its words in each cluster; chunks are compared by cosine. We tried five changes that can be switched on alone: stopword removal, soft assignment to the three nearest clusters, word IDF, bin IDF, and the square root of the histogram.

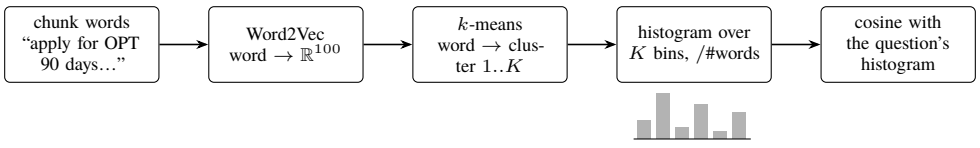


Figure 8: The word-cluster histogram embedding.

Benchmark: for 200 sampled chunks per corpus, `gpt-4o-mini` wrote a paraphrased question each chunk answers; Recall@k is the share of questions whose chunk ranks in the top k among all chunks (Table 2).

	ADS documentation			F-1 regulations		
Embedding	R@1	R@5	MRR	R@1	R@5	MRR
OpenAI `text-embedding-3-small` (1536-d)	0.665	0.925	0.779	0.520	0.905	0.682
BM25 (k_1=1.5, b=0.75)	0.625	0.895	0.738	0.345	0.645	0.484
Mean Word2Vec vector (100-d)	0.425	0.715	0.550	0.215	0.465	0.337
Histogram, baseline (K=200)	0.180	0.405	0.292	0.115	0.225	0.180
+ stopword removal	0.205	0.465	0.331	0.090	0.265	0.184
+ soft assignment	0.260	0.480	0.368	0.115	0.295	0.205
+ word IDF	0.210	0.395	0.311	0.125	0.280	0.202
+ bin IDF	0.285	0.500	0.387	0.110	0.330	0.224
+ square root	0.270	0.510	0.387	0.115	0.325	0.220
all five	0.285	0.600	0.430	0.150	0.345	0.250
all but word IDF, K=800	0.420	0.780	0.576	0.185	0.500	0.325
all but word IDF, K=1600	0.475	0.805	0.629	0.210	0.555	0.361
all but word IDF, K=3200	0.505	0.835	0.656	0.250	0.585	0.396

Table 2: Retrieval of the source chunk for 200 generated questions per corpus, among 1,203 (ADS) and 1,763 (F-1) chunks. Histogram rows: $N = 100$; changes added one at a time to the baseline, then combined.

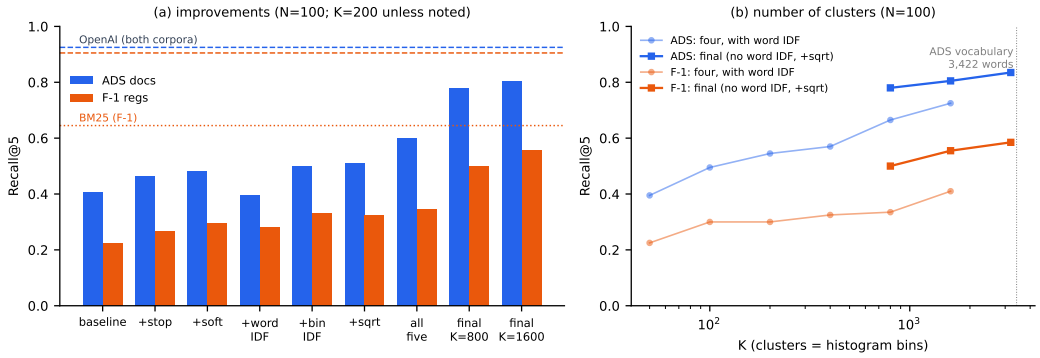


Figure 9: Recall@5 of the histogram embedding. (a) Each change alone at $K = 200$, all five, and the final configuration (all but word IDF) at $K = 800$ and 1,600; dashed: OpenAI, dotted: BM25 on F-1. (b) Effect of K.

Results. The baseline finds the chunk in the top 5 for 41% (ADS) and 23% (F-1) of questions (Figure 9a). At $K = 200$ a cluster holds about 17 unrelated words ("the, at, each, pin") that dominate the histogram; bin IDF, square root and soft assignment each raise ADS Recall@5 to 0.48–0.51, while word IDF hurts and is left out. The number of clusters matters most (Figure 9b): 0.60 at $K = 200$, 0.78 at 800, 0.84 at 3,200, twice the baseline but still below BM25 (0.90), since near the vocabulary size the histogram becomes a bag of words. OpenAI leads on both corpora (0.93, 0.91). On F-1, questions say "work off campus" where pages say "off-campus employment authorization": BM25 falls to 0.65 and the best histogram to 0.59, because Word2Vec on 107,000 words does not learn the paraphrase. Still, in Part 4 the histogram passes all 13 questions: real questions name the terms (OPT, CPT, I-983).